2.2 Cellular Access Technologies

Multiplexing and Multiple Access

Going back to our analogies with radio stations, let us consider the case of a radio station in a locality A. Let's say it's popular, and therefore has loyal customers in localities B and C; and not just because of the good music. Customers in locality B can tune into the station to get news and information that is relevant only to them; ditto for locality C. To achieve this, our radio station sends one signal to users in B and an entirely different signal to users in C, at the same time. How? By using a neat concept called multiplexing. It sends the signal for B on one frequency and C on another. This is called *Frequency Division Multiplexing (FDM)*. FDM, however, is not the only way to multiplex. Another idea for multiplexing is *Time Division Multiplexing (TDM)* - in a span of one second (called the 'Time Frame'), the signal for B can be sent for the first half-second, and that for C can be sent for the second half-second. This repeats every second. TDM is quite popular because you don't need to use different frequencies for different signals.

Armed with this knowledge, we can now apply this to a mobile telephone network. A good mobile phone network allows many callers at the same time. Because callers access the network when needed rather than be connected to it all the time, the term 'Multiplexing' is now replaced by 'Multiple Access'.

A network that separates callers by different frequencies uses *Frequency Division Multiple Access (FDMA)*, one that separates callers by assigning them different time slots uses *Time Division Multiple Access (TDMA)*; and one that separates callers by assigning them unique random codes is called *Code Division Multiple Access (CDMA)*.

In the picture on the next page are four different conversations that need to take place over the cellular network. As we move along, we shall see how each method of multiple access handles these conversations.